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Unlocking Societal Trends in Aadhaar Enrolment and Updates

Case Study: Pattern & Anomaly Detection

Introduction

Our work in this case study analyses the **various patterns** in the Aadhar datasets that includes updates in biometry, demography and new enrolments in the database provided from a duration of **1st March 2025 to 29th December 2025**. This data has features like state, district, pincode and the counts of updates in the age of 5-17 and the age of 17+.

Our objective is to **find any leading impacts**, applications and **irregularities** from performing this exploratory study that can directly or indeed affect the current state of the nation by **reducing identity-related fraud** and improving the Aadhar service in heavy load months or states, and by recognizing the **new influxes of Aadhar** updates in **isolated parts of India** and in low-population regions.

Datasets Used

Our case study **only uses the provided** Aadhar enrollment, biometric and demographic update datasets to **reduce noise** in analysis and **explore unbiased relationships** between the given datasets.

Methodology

Our work in finding patterns, trends and anomalies in the provided datasets include multiple rigorous steps such as cleaning the dataset, processing the data for more accurate and normal representation, performing exploratory data analysis and comparisons of features and graphics with external datasets. The steps are outlined as follows:

Dataset Cleaning

The three provided datasets were classified under three categories:

- Aadhar Enrolment Data
- Aadhar Demographic Update Data
- Aadhar Biometric Update Data

During the data gathering phase, our team noticed that the API data was broken down into multiple files due to the large size of the dataset of **each category**. To improve the convenience of analysis and collate all of the data categories into their respective dataset files, we ran a simple code to **concatenate** all the fragments of the dataset into three singular CSV files for the three categories.

This is an important step towards ensuring that we do not miss any data during analysis or such that there may not be any inconsistencies in the results due to data being broken into separate files.

```python
import os
import pandas as pd


categories = ["biometric", "demographic", "enrolment"]
folder_path = f"./data/api_data_aadhar_"
final_data_path = "./data/datasets"
for cat in categories:
    path = folder_path + cat
    dfs = []
    for f in os.listdir(path):
        df = pd.read_csv(os.path.join(path, f))
        dfs.append(df)
```

```
final = pd.concat(dfs, ignore_index=True)
final.to_csv(f"{final_data_path}/{cat}.csv", index=False)
```

After running the above code cell, we got our collated datasets that require further cleaning. An important step after this is to check whether the datasets contain accurate data. One way to do this is to see how many unique "**state**" values the data contains.

On further inspection, it was revealed that many states were input in differing formats which may cause inconsistencies in final results. For example, there were different records that used "**West Bengal**" and "**WESTBENGAL**" as states. This difference in formatting of the state caused rows to not aggregate properly. We fix this by stripping formatting of all states and creating a canonical mapping to ensure a consistent format of states.

With this step, we were almost done cleaning the datasets. One major issue remained however, that is, the duplicate rows. As the feature "**pincode**" can functionally determine other features like "state" and "district", we form a candidate key using "**pincode**" and "**date**". These two features can uniquely identify any row. We drop all duplicates in the dataset (i.e. having the same values for every feature of the row) and we aggregate by grouping with "date" and "pincode" values by **summing up the counts**. Below is the attached code cell.

```python
df = df.drop_duplicates(keep='first')
df = (
    df.groupby(["date", "pincode"], as_index=False)
      .agg({
          "state": "first",
          "district": "first",
          "bio_age_5_17": "sum",
          "bio_age_17_": "sum",}))
```

Dataset	Rows (Before)	Rows (After)	Rows Removed	% Reduction
Enrolment	1,006,007	868,855	137,152	13.63%
Demographic	2,071,687	1,298,158	773,529	37.33%
Biometric	1,861,108	1,408,962	452,146	24.30%

After this step the data is clean for further analysis and patterns exploration. We perform some simple data preprocessing after this phase.

Data Preprocessing

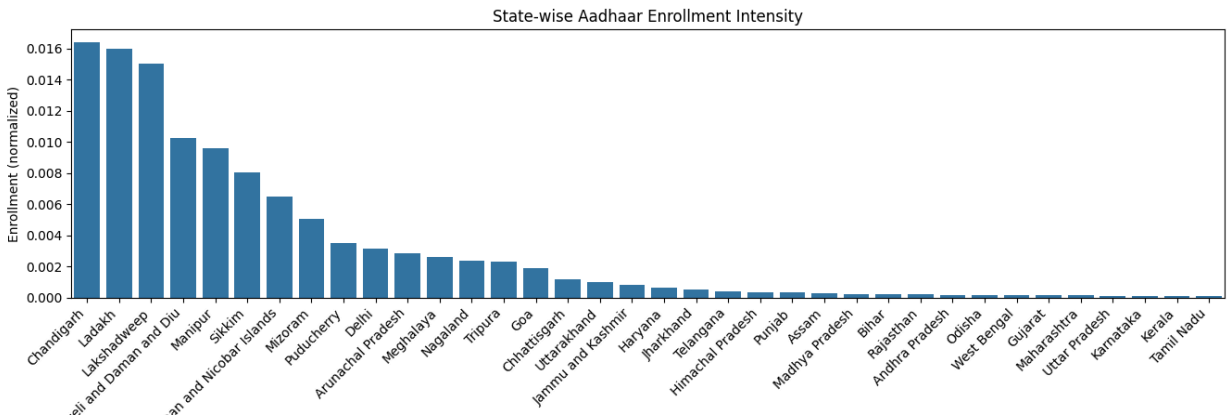
We convert the feature "**date**" from a generic Python object to a datatype of "**datetime64**" as recognized by the "**pandas**" library for easier statistical inference.

After performing some simple checks like looking for further duplicate rows, checking for null values or incorrect data values (like negative counts or incorrect dates), we were not able to find any inconsistent data of such irregularities. Our team decided that the dataset is clean and appropriate for EDA and pattern detection.

We further **normalized** the data using the population of the states to get a more accurate representation and to help notice any irregularities.

```python
df["normalized_enrollment"] = df["total_enrollment"] / df["population"]
df["normalized_enrollment_norm"] = (
    df["normalized_enrollment"] - df["normalized_enrollment"].min()
) / (
    df["normalized_enrollment"].max() - df["normalized_enrollment"].min()
)
state_enrollment = (
    df.groupby("state", as_index=False)
      .agg(
```

```
            normalized_enrollment_norm=("normalized_enrollment_norm", "mean")
        )
).sort_values("normalized_enrollment_norm",ascending=False)
```



Normalization with population helps us to find out

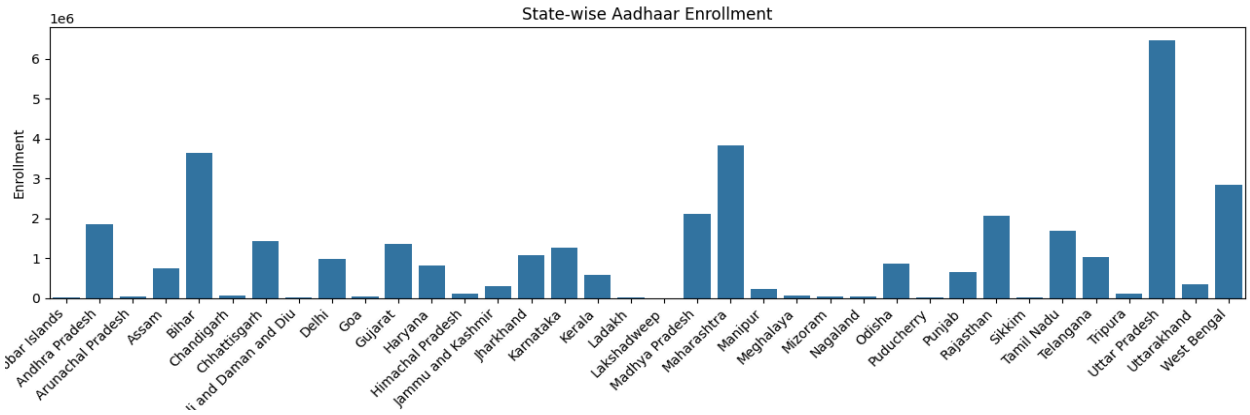
Summary of our data cleaning process:

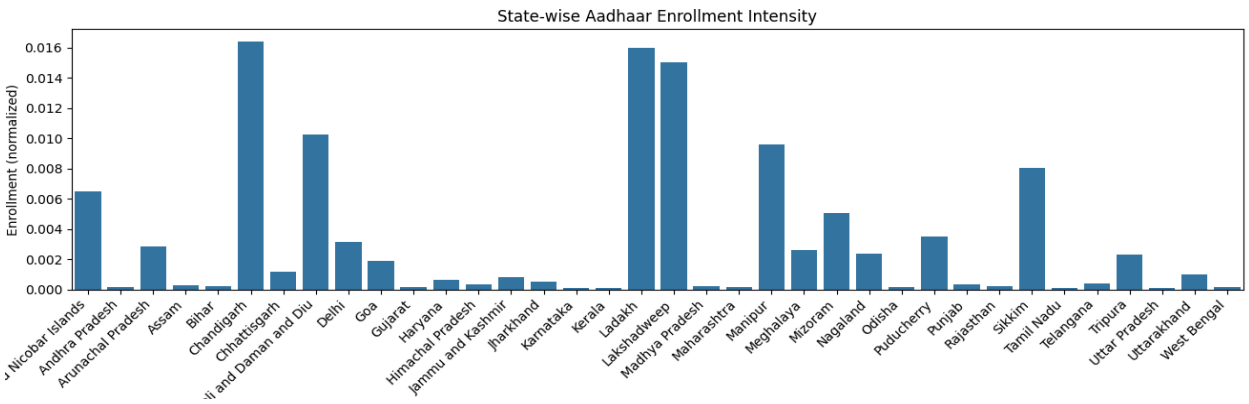
1. Concatenation of fragmented data
2. Formatting of states (and union territories)
3. Duplicate removal and aggregation
4. Normalization of counts across states
5. Final checks (e.g. null values, negative data)

Statistical Inferences

After normalizing by population, Aadhaar enrolment activity is concentrated disproportionately in **small-population regions**, while large states show uniformly low per-capita activity.

If we compare the graphs of the enrollment data when the counts are not normalized and not sorted. The **biometric and demographic updates dataset** exhibit similar results.





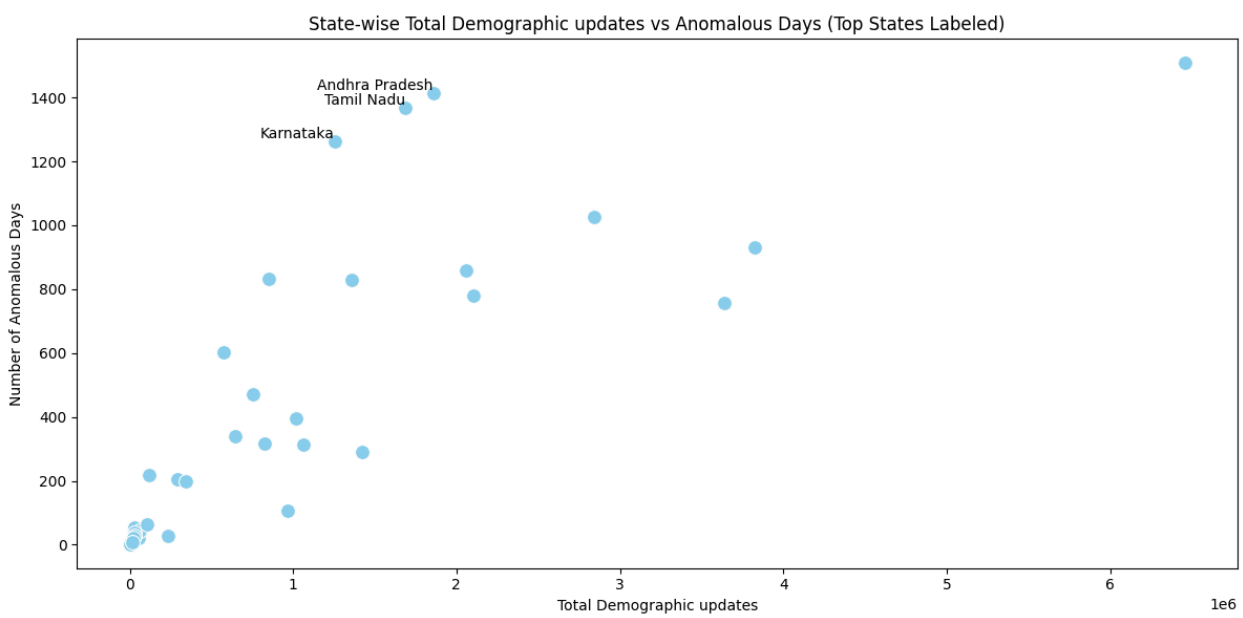
This data comparison shows that the states and union territories that are more isolated (like island states) or having **low population density** (and may be new towards introducing the local people to Aadhar enrollment) even after normalization of the dataset with populations. And larger states with bigger populations having low intensity signifies that new Aadhar enrollments are rarer, pointing towards the fact that a **majority portion of the population of that state already holds an Aadhar card**.

Z - Analysis of Demographic Dataset

Python

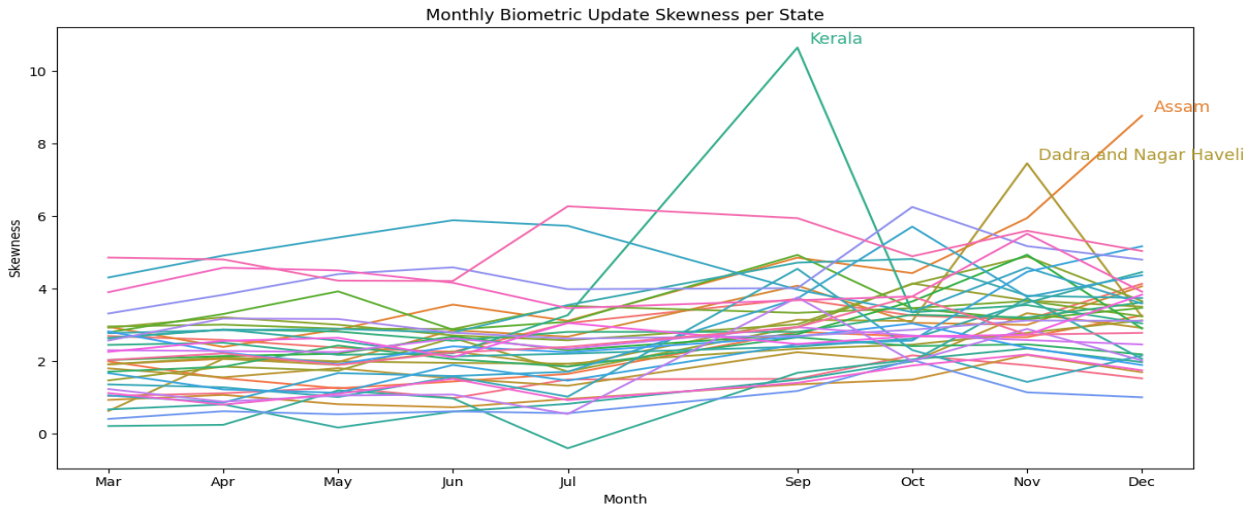
```python
df["rolling_mean"] = df.groupby("pincode")["total_enrollment"].transform(lambda x: x.rolling(7,min_periods=1).mean())

df["z_score"] = ((df["total_enrollment"] - df["rolling_mean"])/df.groupby("pincode")["total_enrollment"].transform("std"))

df["anomaly"] = df["z_score"].abs() > 3
```

```
df["anomaly"] = df["anomaly"].astype(int)

state_anomalies = df.groupby("state")["anomaly"].sum().sort_values()

df["anomaly_flag"] = (df["z_score"].abs() > 3).astype(int)

state_summary = df.groupby("state").agg(

    total_enrollment=("total_enrollment", "sum"),

    anomalies=("anomaly_flag", "sum")

).reset_index()

highest_enrollment_state =
state_summary.loc[state_summary["total_enrollment"].idxmax(), "state"]

top_states = (

    state_summary[state_summary["state"] != highest_enrollment_state]

    .nlargest(3, "anomalies")

)
```

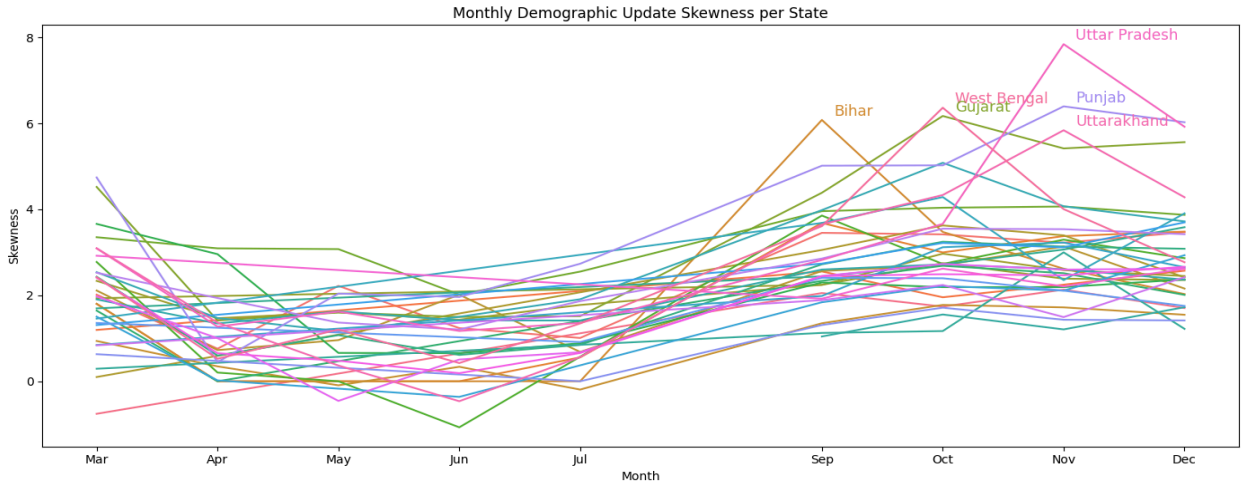


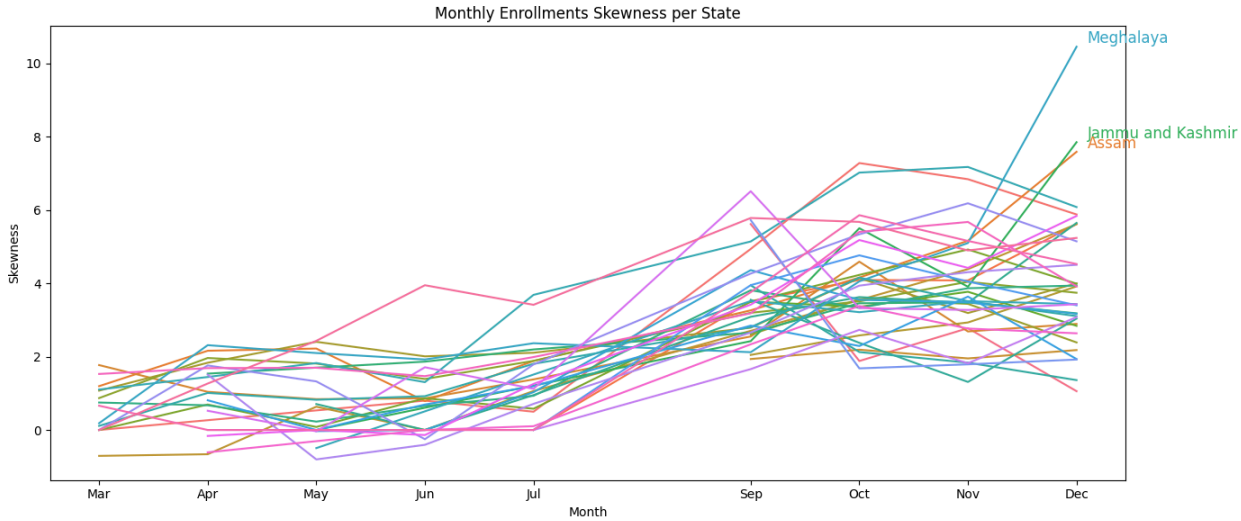
On the basis of Z-Analysis of the demographic data that uses the "**rolling mean**" of a time period of 7 days across the states to find out any anomalies that point towards a sudden influx in Aadhar demographic updates, we find that states like **Andhra Pradesh, Karnataka and Tamil Nadu** show higher than average number of anomalous days across all dates.

Anomalies in Skew Data (Multivariate analysis)



Note how during the later months of the year near **September** and **November** there is a sudden skew spike in biometric update activity of some states, notably **Assam**, **Kerala** and **Dadra and Nagar Haveli and Daman and Diu**. Other datasets follow a similar skew pattern.



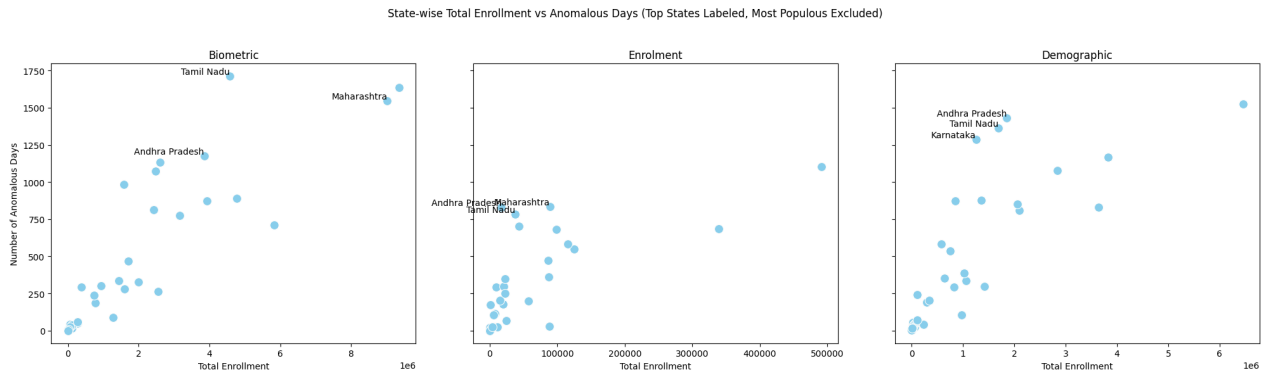


Here during the final months of the year there is an increased enrollment count and **Meghalaya** has an unusually suspicious increase in activity.

Attached is the code cell used to create the above skew plots:

```python
from scipy.stats import skew
import calendar
df['month'] = df['date'].dt.month
monthly_skew = (
    df.groupby(["state", "month"])["total_enrollment"]
      .apply(lambda x: skew(x.dropna()))
      .reset_index(name="skewness")
)
monthly_skew_filtered = monthly_skew[monthly_skew["month"] >= 3]
peak_skew_per_state = monthly_skew_filtered.groupby("state")["skewness"].max()
top_states = peak_skew_per_state.sort_values(ascending=False).head(3).index
```

Anomalies in Standard Score (Z-Score)



We observe the various scatter plots obtained for our different datasets in the inference for the number of anomalous days by comparison with the standard deviation (which is known as the Z score method). We define "**anomalous days**" as the measure of how many standard deviations a state is from the mean count in date, indicating if a value is unusual. This is a **trivariate analysis**.

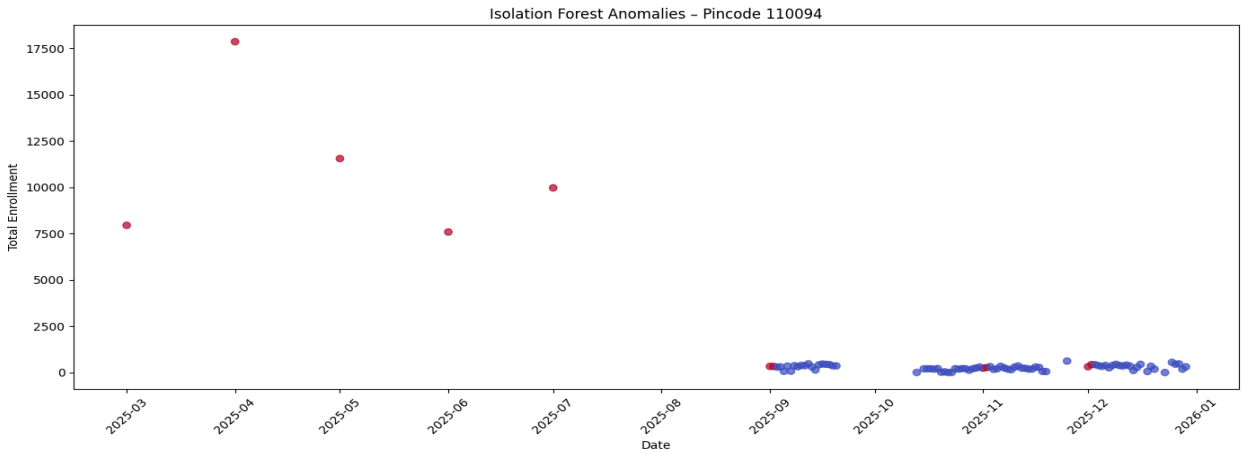
In the Biometric update data, Tamil Nadu has an extremely high indication in updates. This may point to **fraudulent activity** in Tamil Nadu related to biometric identity verification measures.

The Enrollment data shows **small deviation in anomalies**, though notable states include Andhra Pradesh, Maharashtra and Tamil Nadu. Similar results are exhibited in the Demographic update data, with the suspicious recurrence of Tamil Nadu once more.

Note that we **ignore outliers** like Uttar Pradesh (top right point in all plots), since the data we have for it is comparatively massive in size relative to other states.

Machine Learning

On experimenting with unsupervised machine learning models like **K-Means Clustering** and **Isolation Forest**, we observe that the models flag higher enrollment and update counts as anomalous which is not a valid indicator of a meaningful pattern in the dataset.



Using a sample set of a random pincode, it can be noted that the model exhibits the same properties here as listed above. Even if we introduce temporal data (like rolling mean), it produces similar results. Therefore, we focus on **statistical analyses** and base our findings upon that.

Insights, Impacts and Applications

Our work explored multiple perspectives in the given datasets, and in our attempt to find meaningful patterns, we present our insights and propose impacts and solutions.

- A noticeable spike in the **later months** of the year showed increased update counts and enrollments, which might help improve **load balancing** in online servers and **crowd management** at UIDAI centers.
- A recurring pattern observed in our study provides us with measures of counts for **low-population density areas and isolated territories**. Improving Aadhar awareness in those regions would be beneficial.
- **Northern states** show high migration rates due to spikes in **demographic changes** which comes out to be an interesting case study for future exploration.
- Many **southern states** show **anomalous trends** occurring in the biometric update data or sudden changes in enrollment patterns. This may be a cause for concern and should be explored further.